



## Chapter 6, Frequency Response and Bode Plots

One of the major branches of electrical engineering is signal processing. Signal processing is concerned with manipulating signals to extract information. Signal is a voltage or current that contains some information. For example, electrocardiogram (ECG) is an electric signal generated by human heart. Using circuits and computers one can extract useful information about the health condition of the patient's heart.

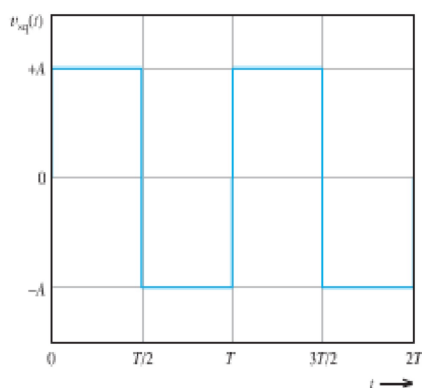
Previously we have learned to analyze circuits that are excited by either a dc or a sinusoidal source.

However, most real world electric signals are not sinusoidal. How can we analyze circuits excited by non-sinusoidal sources? How can we express non-sinusoidal signals by phasors?

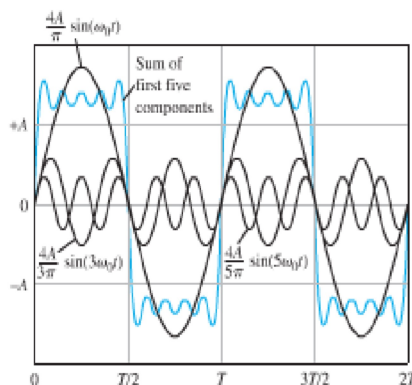
Joseph Fourier (1768-1830) developed a mathematical technique to express the non-sinusoidal functions as the sum of a set of sinusoidal components having various frequencies, amplitudes and phases.

For example using Fourier analysis, a square wave has the following components:

$$v_{sq}(t) = \frac{4A}{\pi} \sin(\omega_0 t) + \frac{4A}{3\pi} \sin(3\omega_0 t) + \frac{4A}{5\pi} \sin(5\omega_0 t) + \dots$$

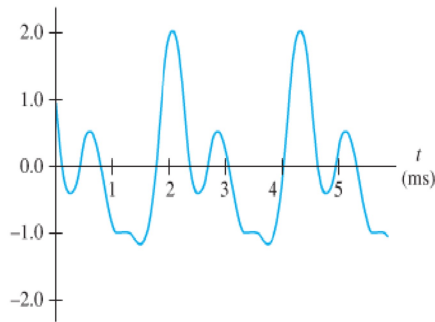


(a) Periodic square wave

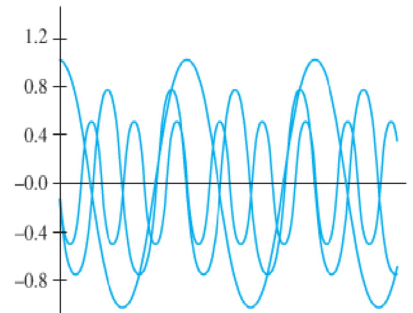


(b) Several of the sinusoidal components and the sum of the first five components

or, the components of this music waveform are:



(a) Music waveform

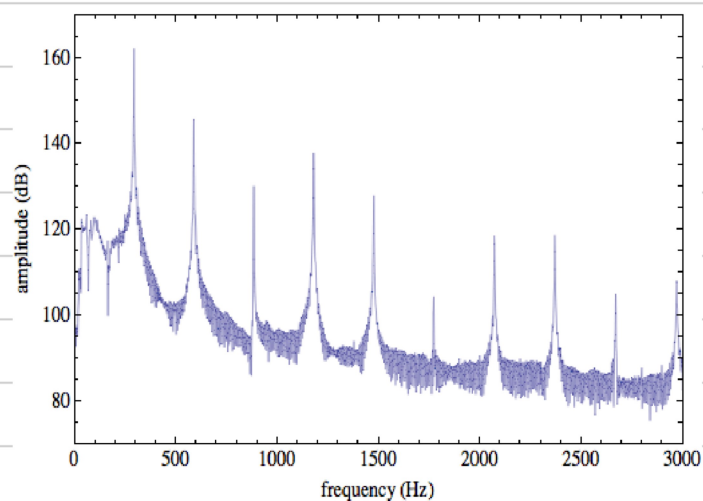


(b) Sinusoidal components

### Frequency Ranges of Selected Signals

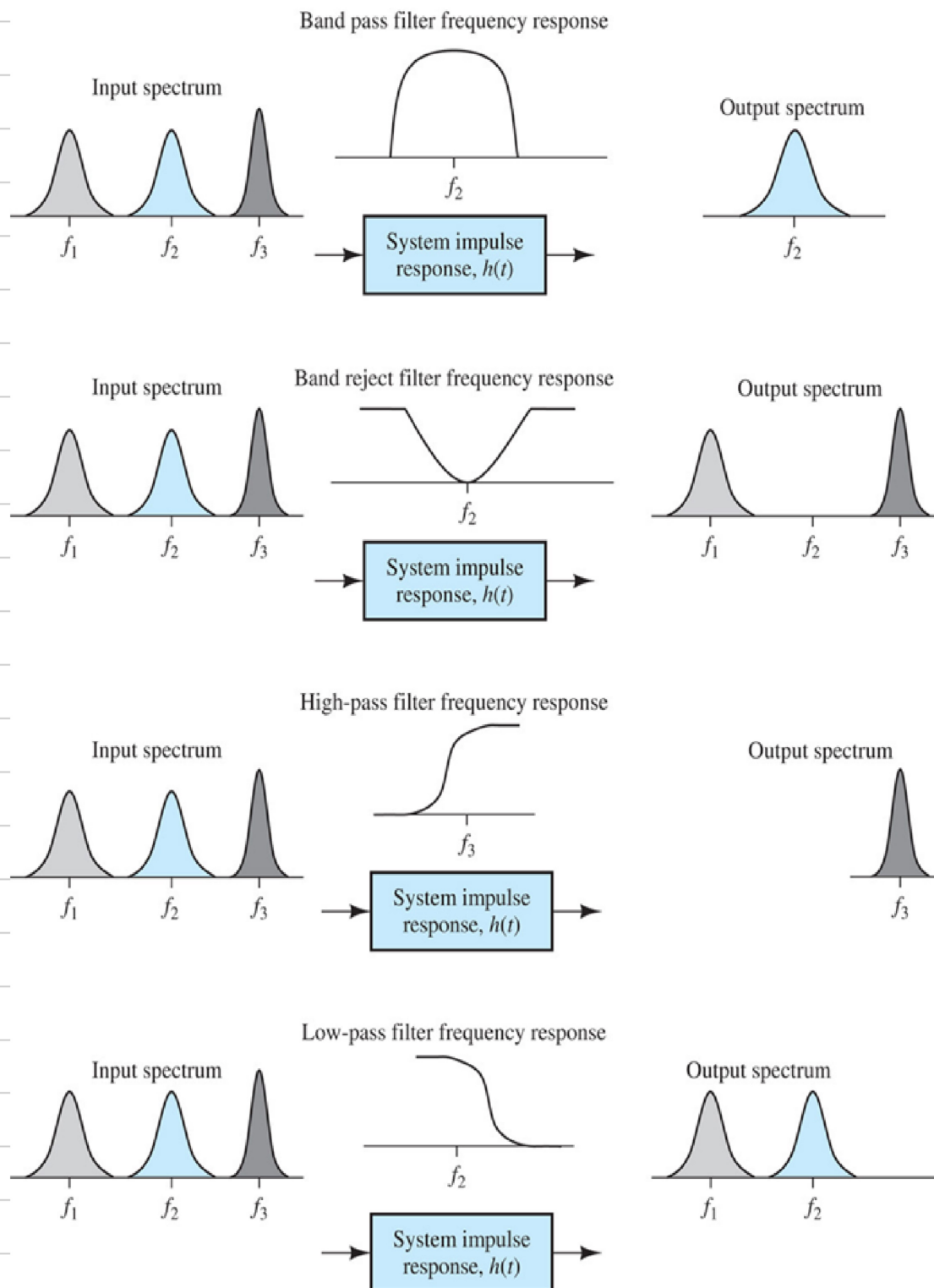
|                                         |                                     |
|-----------------------------------------|-------------------------------------|
| Electrocardiogram                       | 0.05 to 100 Hz                      |
| Audible sounds                          | 20 Hz to 15 kHz                     |
| AM radio broadcasting                   | 540 to 1600 kHz                     |
| Analog video signals (U.S. standards)   | Dc to 4.2 MHz                       |
| FM radio broadcasting                   | 88 to 108 MHz                       |
| Cellular phone                          | 824 to 894 MHz and 1850 to 1990 MHz |
| Satellite television downlinks (C-band) | 3.7 to 4.2 GHz                      |
| Digital satellite television            | 12.2 to 12.7 GHz                    |

Spectrum analyzer is a laboratory instrument used to determine the frequencies, amplitude and phases of a signal.



## Filters

In some applications we want to select a certain range of frequencies and discard the rest. This can be done using electric circuits called filters.



Some of the frequency components of the input signal



are passed to the output, and the others are rejected.

## Transfer Function

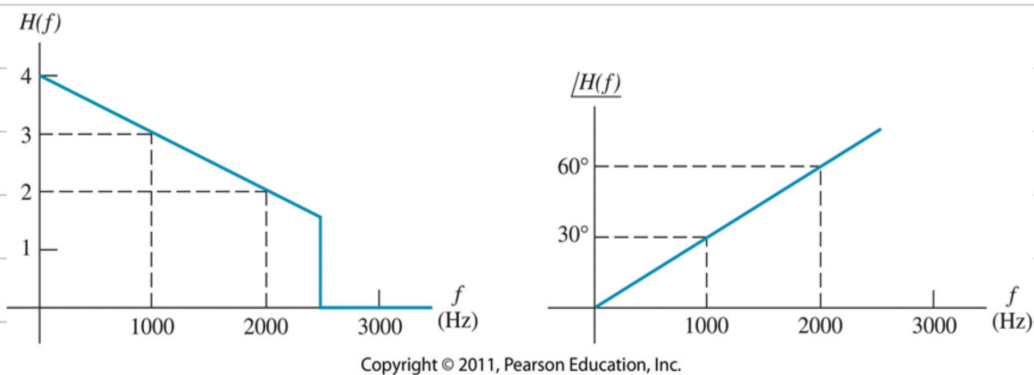
Since the impedance of an inductor or capacitor is a function of frequency, the ratio of the output to input is a function of frequency. The ratio of phasor  $V_{out}$  to phasor  $V_{in}$  is called transfer function.

$$H(f) = \frac{V_{out}}{V_{in}} = \frac{|V_{out}| \angle \theta_{out}}{|V_{in}| \angle \theta_{in}} = \frac{|V_{out}|}{|V_{in}|} \angle \theta_{out} - \theta_{in}$$

Transfer function is a complex quantity having both magnitude and phase angle.

$$|H(f)| = \frac{|V_{out}|}{|V_{in}|} \quad \text{and} \quad \angle H(f) = \theta_{out} - \theta_{in}$$

**Ex.** The magnitude and phase of the transfer function of a filter is shown. If  $v_{in}(t) = 2 \cos ( 2\pi \cdot 2000 t)$ , find the expression for  $v_{out}(t)$ .



The only frequency component in  $V_{in}$  is 2000 Hz. At this frequency the magnitude of the output is twice the input and the phase angle is shifted by  $60^\circ$  at the output. Therefore,

$$V_{in}(t) = 2 \cos(2\pi \times 2000t) \Rightarrow V_o(t) = 4 \cos(2\pi \times 2000t + 60^\circ)$$

If we apply  $v_{in}(t) = 2 \cos(2\pi \cdot 1000 t) + 3 \cos(2\pi \cdot 3000 t)$  to the input of the same filter what do we expect to see at the output?

$$\text{At } f = 1000 : |H| = 3 \text{ and } \angle H = 30^\circ$$

$$\text{At } f = 3000 : |H| = 0$$

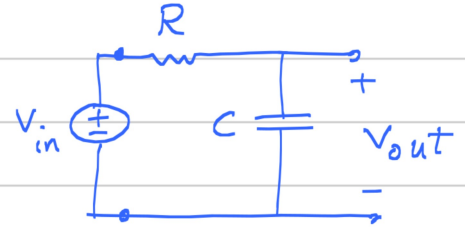
$$\text{Therefore, } V_{out}(t) = 6 \cos(2\pi \cdot 1000 t + 30^\circ)$$

The graphic equalizer of an audio system has an adjustable transfer function. It can be adjusted to achieve the most pleasing sound.

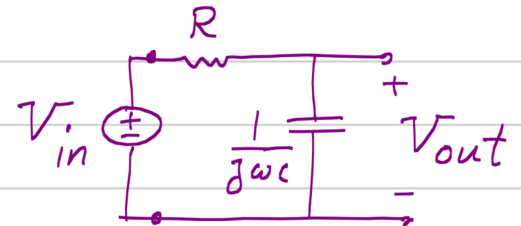
## First order RC low pass filters

A low pass filter tends to pass the low frequency components and reject the high frequency components.

$$V_{out} = V_{in} \frac{\frac{1}{j2\pi fC}}{\frac{1}{j2\pi fC} + R} = V_{in} \frac{1}{1 + j2\pi fRC}$$



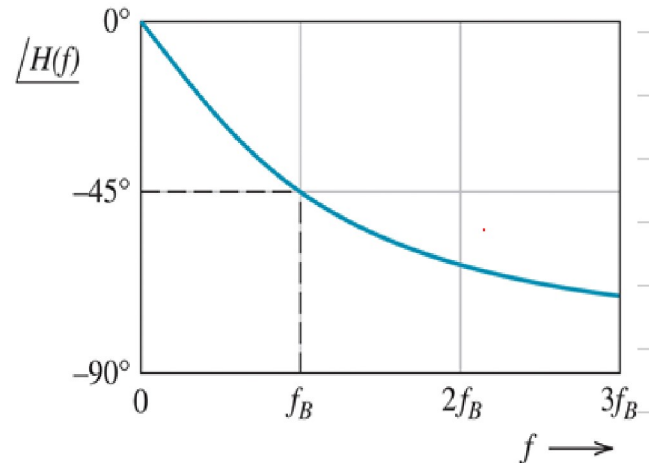
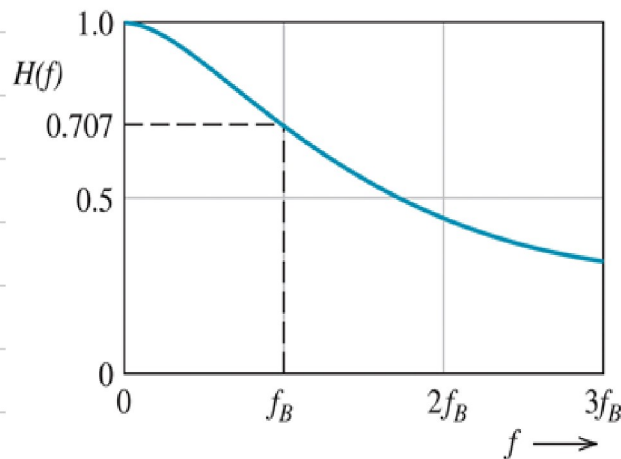
$$H(f) = \frac{V_{out}}{V_{in}} = \frac{1}{1 + j2\pi fRC}$$



define  $f_B = \frac{1}{2\pi RC}$  (Break freq., Corner freq., or cut off freq.)

$$H(f) = \frac{1}{1 + j \frac{f}{f_B}}$$

$$|H(f)| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_B}\right)^2}}, \quad \angle H(f) = -\tan^{-1} \frac{f}{f_B}$$



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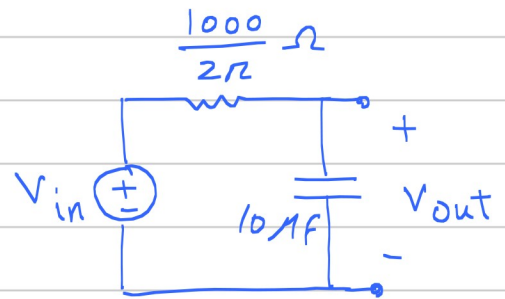
The break frequency  $f_B$  is also called half-power frequency because at this frequency the voltage of the output is  $\frac{1}{\sqrt{2}} V_{in}$  and the power is one half of the input power.

**Ex.** The input signal to the low pass RC filter is  $v_{in}(t) = 5 \cos(20\pi t) + 50 \cos(200\pi t)$ .

Find the expression for the output signal.

cut off frequency;

$$f_B = \frac{1}{2\pi RC} = \frac{1}{2\pi \left(\frac{1000}{2\pi}\right)(10 \times 10^{-6})} = 100 \text{ Hz}$$



There are two freq. components at the input:

$$f_1 = 10 \text{ Hz} \quad \text{and} \quad f_2 = 100 \text{ Hz}$$

$$H(f) = \frac{1}{\sqrt{1 + \left(\frac{f}{f_B}\right)^2}} \angle -\tan^{-1} \frac{f}{f_B}$$

$$\text{at } f = 10 \text{ Hz} : H_{(f=10)} = \frac{1}{\sqrt{1 + \left(\frac{10}{100}\right)^2}} \angle -\tan^{-1} \frac{10}{100} = 0.995 \angle -5.7^\circ$$

$$\Rightarrow V_o(f=10) = H_{(f=10)} \cdot V_{in}(f=10) = (0.995 \angle -5.7^\circ)(5 \angle 0^\circ)$$

$$V_o(f=10) = 4.97 \angle -5.71^\circ$$

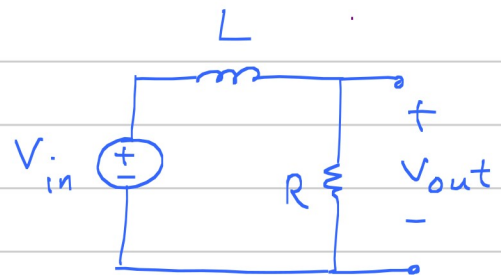
$$\text{at } f = 100 \text{ Hz} : H_{(f=100)} = \frac{1}{\sqrt{1 + \left(\frac{100}{100}\right)^2}} \angle -\tan^{-1} \frac{100}{100} = 0.707 \angle -45^\circ$$

$$\Rightarrow V_o(f=100) = (0.707 \angle -45^\circ)(50 \angle 0^\circ) = 35.35 \angle -45^\circ$$

$$V_o(t) = 4.97 \cos(20\pi t - 5.71^\circ) + 35.35 \cos(200\pi t - 45^\circ)$$

## First order RL low pass filters

$$V_{out} = \frac{R}{R + j2\pi fL} V_{in}$$



$$H(f) = \frac{1}{1 + j2\pi f \frac{L}{R}} = \frac{1}{1 + j \frac{f}{f_B}}, \quad f_B = \frac{R}{2\pi L}$$

## Decibels and logarithmic scales

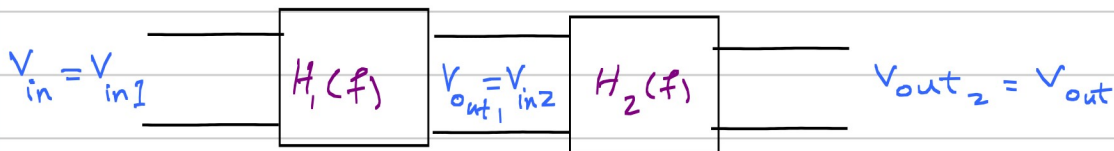
The magnitude of transfer function is usually expressed in logarithmic scale or decibels.

$$|H(f)|_{dB} = 20 \log |H(f)|$$

The advantage of using a logarithmic scale is that the variations in the magnitude and frequency in the low range as well as the variations in the high range can be clearly shown on a single plot.

## Cascaded network

When the output of one system having transfer function  $H_1$  is connected to the input of another system with transfer function  $H_2$ , the transfer function of the two system combined is  $H = H_1 H_2$ .



$$V_{out} = H_2(f) \cdot V_{in2} = H_2(f) \cdot (H_1(f) \cdot V_{in})$$

$$\Rightarrow H(f) = \frac{V_{out}}{V_{in}} = H_1(f) \cdot H_2(f)$$

$$|H(f)|_{dB} = 20 \log |H(f)| = 20 \log [H_1(f) \cdot H_2(f)]$$

$$= 20 \log |H_1(f)| + 20 \log |H_2(f)|$$

$$= |H_1(f)|_{dB} + |H_2(f)|_{dB}$$